

Endovascular Today - US Device Guide - Mechanical Thrombectomy (Neurovascular)

Company Name	Product Name	Sheath Compatibility (F)	Guidewire Compatibility (inch)	Working Length (cm)	Mode of Operation	US FDA Indicated Use
Cerenovus	Embotrap III Revascularization Device - 22 mm (indication 1.5-5mm)	–	–	194 (overall length)	–	Designed for use in the anterior and posterior neurovasculature in vessels such as the internal carotid artery, the M1 or M2 segments of the middle cerebral artery, and basilar arteries (vessel sizes ranging from 1.5 mm up to 5 mm for the 5-mm diameter device)
Cerenovus	Embotrap III Revascularization Device - 37 mm (indication 1.5-5 mm)	–	–	195 (overall length)	–	Designed for use in the anterior and posterior neurovasculature in vessels such as the internal carotid artery, the M1 or M2 segments of the middle cerebral artery, and basilar arteries (vessel sizes ranging from 1.5 mm up to 5 mm for the 5-mm diameter device)
Cerenovus	Embotrap III Revascularization Device - 45 mm (indication 1.5-6.5mm)	–	–	196 (overall length)	–	Designed for use in the anterior and posterior neurovasculature in vessels such as the internal carotid artery, the M1 or M2 segments of the middle cerebral artery, and basilar arteries (vessel sizes ranging from 1.5 mm up to 6.5 mm for the 6.5-mm diameter device)
Imperative Care (Stroke)	Zoom 6F Insert Catheters - Zoom VTK, Zoom VRT, Zoom SIM	–	–	137–143	The first and only 6-F Insert Catheter Portfolio with three shapes designed specifically for stroke patients	The Zoom 6F Insert Catheters are indicated for use in delivering radiopaque media to selected sites in the peripheral vascular system in conjunction with routine diagnostic procedures
Imperative Care (Stroke)	Zoom Aspiration Catheters: Zoom 71, Zoom 55, Zoom 45, Zoom 35	–	–	137–160	Vessel-matched aspiration catheters with angled TRX Tip; designed to be used as a complete system with the Zoom 88 Large Distal Platform, Zoom POD, and Zoom Pump	The Zoom 88 large distal platform (Zoom 88) is indicated for the introduction of interventional devices into the peripheral, coronary, and neurovasculature; Zoom 88 should only be used by physicians who have received appropriate training in interventional techniques and are proficient in using guide sheaths and catheters; the Zoom aspiration catheters (Zoom 71, Zoom 55, Zoom 45, Zoom 35), with the Zoom aspiration tubing set and Zoom aspiration pump (or equivalent vacuum pump), are indicated for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral – M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset; patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment; the Zoom aspiration pump is intended for general suction use in hospitals and clinics; the Zoom aspiration tubing set is intended to connect the Zoom aspiration catheter to the Zoom canister of the Zoom aspiration pump and to allow the user to control the fluid flow; the Zoom aspiration catheters, Zoom aspiration pump, and Zoom aspiration tubing set should only be used by physicians who have received appropriate training in interventional techniques and treatment of acute ischemic stroke
Medtronic	Solitaire Platinum Revascularization	8, 9 (balloon guide)	0.014	180 (Solitaire push wire)	Designed to restore blood flow in patients experiencing ischemic	Indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in

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	Device 4 X 20, 4 X 40, 6 X 20				stroke due to large intracranial vessel occlusion and for use in the neurovasculature, such as the internal carotid artery, M1 and M2 segments of the middle cerebral artery, basilar, and the vertebral arteries	patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received IV tPA; endovascular therapy with the device should be started within 6 hours of symptom onset; indicated to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for IV tPA or who fail IV tPA therapy are candidates for treatment
Medtronic	Solitaire X Revascularization Device (4 X 20; 4 X 40; 6 X 20; 6 X 24; 6 X 40 mm)	–	0.021	200	–	(1) Indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA)—endovascular therapy with the device should be started within 6 hours of symptom onset; (2) indicated to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset—patients who are ineligible for IV t-PA or who fail IV t-PA therapy are candidates for treatment; (3) indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion of the internal carotid artery (ICA) or middle cerebral artery (MCA)-M1 segments with smaller core infarcts (< 70 mL by CTA or MRA; < 25 mL by MR-DWI)—endovascular therapy with the device should start within 6–16 hours of time last seen well in patients who are ineligible for IV t-PA or who fail IV t-PA therapy
Medtronic	Solitaire X-Pack (2 pack: 4 X 40 or 6 X 40 mm with Phenom 21 or 27)	–	–	200	–	(1) Indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA)—endovascular therapy with the device should be started within 6 hours of symptom onset; (2) indicated to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset—patients who are ineligible for IV t-PA or who fail IV t-PA therapy are candidates for treatment; (3) indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion of the internal carotid artery (ICA) or middle cerebral artery (MCA)-M1 segments with smaller core infarcts (< 70 mL by CTA or MRA; < 25 mL by MR-DWI)—endovascular therapy with the device should start within 6–16 hours of time last seen well

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Medtronic	Solitaire X-Pack (3 pack: 4 X 40 or 6 X 40 mm with Phenom 21 or 27 or React 68 or 71)	–	–	200	–	in patients who are ineligible for IV t-PA or who fail IV t-PA therapy (1) Indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA)—endovascular therapy with the device should be started within 6 hours of symptom onset; (2) indicated to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset—patients who are ineligible for IV t-PA or who fail IV t-PA therapy are candidates for treatment; (3) indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion of the internal carotid artery (ICA) or middle cerebral artery (MCA)-M1 segments with smaller core infarcts (< 70 mL by CTA or MRA; < 25 mL by MR-DWI)—endovascular therapy with the device should start within 6–16 hours of time last seen well in patients who are ineligible for IV t-PA or who fail IV t-PA therapy
Penumbra, Inc. (Neuro)	3D Revascularization Device	6 (long sheath [Neuron MAX])	0.020	200 (delivery wire; compatible with Velocity Delivery microcatheter [160 cm] and 3MAX Reperfusion catheter [160 cm])	Designed for use in combination with Reperfusion catheters; the architecture enables maximum clot capture capability through four intraluminal chambers and is optimized for use with aspiration delivered by ACE catheters or the Penumbra JET 7 Reperfusion catheter	As part of the Penumbra system, the Penumbra 3D Revascularization device is indicated for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral—M1 and M2 segments) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment. As part of the Penumbra system, the Penumbra 3D Revascularization device is indicated for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral—M1 and M2 segments) within 8 hours of symptom onset. Patients who are ineligible for thrombolytic drug therapy or who failed thrombolytic drug therapy are candidates for treatment
Penumbra, Inc. (Neuro)	Penumbra System Reperfusion Catheters: RED 72, RED 72 with SENDit Technology, RED 68, RED 62, RED 43, Penumbra JET D, 4MAX, 3MAX, and 026	6 (RED 72, RED 72 with SENDit Technology, RED 68, RED 62, RED 43, and Penumbra JET D require 8-F short sheath or 6-F long sheath)	0.014–0.024	132–160	Direct aspiration clot removal with Penumbra Engine aspiration source	As part of the Penumbra system, the Reperfusion catheters and separators are indicated for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral—M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment. As part of the Penumbra system, the Reperfusion catheters and separators are indicated for use in the revascularization of patients with acute ischemic stroke secondary to

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						intracranial large vessel occlusive disease (within the internal carotid, middle cerebral—M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for thrombolytic drug therapy or who failed thrombolytic drug therapy are candidates for treatment
Penumbra, Inc. (Neuro)	Penumbra System Separator and Separator Flex: 5MAX, 4MAX, 3MAX, and 026	–	–	175–200 (Separator and Separator Flex length)	Separator-assisted clot debulking, if needed	As part of the Penumbra system, the Reperfusion catheters and separators are indicated for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral—M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment As part of the Penumbra system, the Reperfusion catheters and separators are indicated for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral—M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for thrombolytic drug therapy or who failed thrombolytic drug therapy are candidates for treatment
Rapid Medical	Tigertriever 13	0.0165/0.013 inch (ID)	–	230	Adjusts to the vasculature and clot providing an atraumatic approach; precise control to capture the thrombus and remove the tension from the vasculature before removal leading to safe and effective procedure; the user can control the manual expansion of the braided distal portion into the clot using the handle component of the device	Intended to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA), or who fail IV t-PA therapy, are candidates for treatment
Rapid Medical	Tigertriever 17	0.017 inch (ID)	–	220	Adjusts to the vasculature and clot providing an atraumatic approach; unique operation changes these procedures from a passive to an active user approach that accelerates clot capture and potentially reduces the risk of vascular injury during removal; the user can control the manual expansion of the braided distal portion into the clot using	Intended to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA), or who fail IV t-PA therapy, are candidates for treatment

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					the handle component of the device	
Rapid Medical	Tigertriever 17 Ultra	0.017 inch (ID)	–	220	Adjusts to the vasculature and clot providing an atraumatic approach; unique operation changes these procedures from a passive to an active user approach that accelerates clot capture and potentially reduces the risk of vascular injury during removal; the user can control the manual expansion of the braided distal portion into the clot using the handle component of the device	Intended to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for thrombolytic drug therapy or who failed thrombolytic drug therapy are candidates for treatment
Rapid Medical	Tigertriever 25	0.025 inches	–	220	Adjusts to the vasculature and clot providing an atraumatic approach; unique operation changes these procedures from a passive to an active user approach that accelerates clot capture and potentially reduces the risk of vascular injury during removal; the user can control the manual expansion of the braided distal portion into the clot using the handle component of the device	Intended to restore blood flow by removing thrombus from a large intracranial vessel in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for thrombolytic drug therapy or who failed thrombolytic drug therapy are candidates for treatment
Stryker	Trevo NXT ProVue Retriever (3 X 32 mm, 4 X 28 mm, 4 X 41 mm, 6 X 37 mm)	8, 9 (balloon guide catheter)	0.014	200 (for all stent sizes)	Mechanical thrombectomy, 360° of consistently large cells, a soft distal tip, and full-length radiopacity	(1) Indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA); endovascular therapy with the device should start within 6 hours of symptom onset; (2) intended to restore blood flow in the neurovasculature by removing thrombus in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for IV t-PA or who fail IV t-PA therapy are candidates for treatment; and (3) indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior

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						circulation, large vessel occlusion of the internal carotid artery or middle cerebral artery–M1 segments with smaller core infarcts (0–50 mL for age < 80 years, 0–20 mL for age ≥ 80 years); endovascular therapy with the device should start within 6–24 hours of time last seen well in patients who are ineligible for IV t-PA or who fail IV t-PA therapy
Stryker	Trevo XP ProVue Retriever (3 X 20 mm, 4 X 20 mm, 4 X 30 mm, 6 X 25 mm)	8, 9 (balloon guide catheter)	0.014	190 for 3 X 20 mm, 180 for 4 X 20 mm, 4 X 30 mm, and 6 X 25 mm	Mechanical thrombectomy; 360° of consistently large cells, a soft distal tip and full-length radiopacity	Indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large vessel occlusion, and smaller core infarcts who have first received intravenous tissue plasminogen activator (IV t-PA); endovascular therapy with the device should start within 6 hours of symptom onset; intended to restore blood flow in the neurovasculature by removing thrombus in patients experiencing ischemic stroke within 8 hours of symptom onset; patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment; indicated for use to restore blood flow in the neurovasculature by removing thrombus for the treatment of acute ischemic stroke to reduce disability in patients with a persistent, proximal anterior circulation, large-vessel occlusion of the internal carotid artery (ICA) or middle cerebral artery (MCA)-M1 segments with smaller-core infarcts (0–50 mL for age < 80 years, 0–20 mL for age ≥ 80 years); endovascular therapy with the device should start within 6–24 hours of time last seen well in patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy